

Analysis of the Toric Code

PH5842 Term Paper

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In this report, we look at some of the important properties of topological quantum error correction protocols through the lens of the Toric Code. We analyze the framework presented in [1], and understand some of the properties of the code, its logical states, and operators in the encoded space. We also develop conditions for quantum error correction specific to the code, and present a rigorous error model accounting for both qubit and measurement errors over time that helps map the problem to a well-studied phase transition in statistical physics, thereby allowing us to estimate the threshold single-qubit-error probability below which quantum information can be reliably stored. This provides a quantitative benchmark for fault-tolerant quantum memory.

I. INTRODUCING TORIC CODES

A. Why encode Quantum Information on a torus?

Quantum systems are inherently susceptible to decoherence and operational imperfections, making error correction essential for reliable quantum computation. While initial days of QEC saw many extensions of repetition codes (which was inspired from many of the classical codes), the natural extension to such repetition codes, especially considering the use-case of quantum memory, is to exploit both our ability to control qubits at the individual level as well as the multi-dimensional nature of our *nature* and pack multiple of these qubits together spatially to define an error-resistant logical qubit. Thus a natural way of extending the one-dimensional n-qubit repetition code is a surface code where an ensemble of qubits are arranged on a surface, of which an interesting case is a torus (which can be thought of as a square lattice with periodic boundary conditions).

These codes provide a way to scale the encoding to higher number of qubits easily, while also providing surprisingly good resistance to noise. In this report, we will thus introduce the toric code, discuss the effect of noise and how to tackle it and also describe a mathematical model to estimate the threshold probability for this code.

B. Properties of the Toric Code

The Toric code formally refers to a subspace of the Hilbert space of some physical qubits that reside on a torus. In our analysis of the code, this can be better visualized as being defined on a two-dimensional grid with periodic boundary conditions, where physical qubits occupy the centers of "links" (or edges) of the grid (Figure 1). The Toric code is analyzed using stabilizer formalism, and hence to aid our analysis we

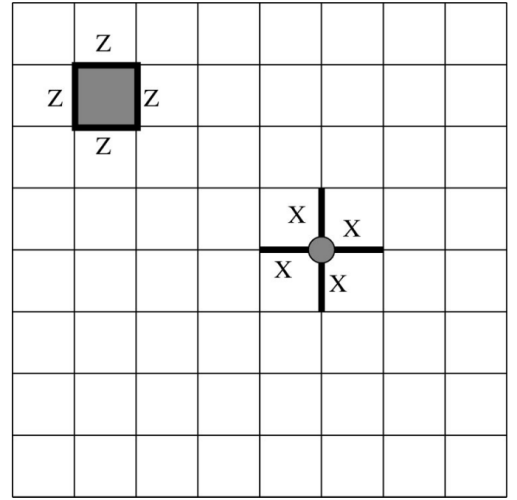


Figure 1. Plaquette and Site operations. Physical qubits reside at the centers of links.

define a set of Stabilizer operators over this 2D grid.

"Plaquette" operators are defined over square unit-cells of the lattice, given by:

$$Z_P = \bigotimes_{\ell \in P} Z_\ell, \quad (1)$$

where P is a plaquette (Figure 1) and $\ell \in P$ denotes an edge bounding the plaquette P . Each Z_ℓ is a Pauli- Z operator acting on the edge ℓ belonging to the plaquette.

On a similar vein, we define a set of site operators, given by

$$X_s = \bigotimes_{\ell \ni s} X_\ell, \quad (2)$$

where each site s is a vertex on the grid, and the product runs over the four edges ℓ that have s as one of the ends. Each X_ℓ is a Pauli- X operator acting on edge ℓ

belonging to the site. Together, they're referred to as the "check" operators (will come to why it is so.)

These specialized set of operators define the Toric code and follow some interesting set of properties:

1. Sites and Plaquettes as Stabilizer generators

Each site operator X_s commutes with both itself as well as each plaquette operator Z_P :

$$[X_s, Z_P] = 0$$

This can be shown via the following arguments:

- Each site (plaquette) operator X_s (Z_P) trivially commutes with every other site (plaquette) operator $X_{s'}$ ($Z_{P'}$), since they're all either an X (Z) operation on the same qubit or on independent qubits.
- If X_s and Z_P act on disjoint qubits (i.e. the site does not belong to the plaquette), they trivially commute.
- If a site (vertex) belongs to a plaquette, then they share exactly two qubits, and each shared qubit contributes a factor of -1 due to the anticommutation of X and Z . Two such factors cancel and the net commutator is 0.

2. Independence of stabilizer generators

On an $L \times L$ lattice on a torus, there are L^2 plaquette operators and L^2 site operators (corresponding to each face and vertex on the lattice, respectively).

However, not all of these stabilizers are independent. Since each edge (qubit) lies on exactly two plaquettes and connects exactly two vertices, the product over all plaquette operators satisfies:

$$\prod_P Z_P = \mathbb{I}^{\otimes n}, \quad \text{and} \quad \prod_s X_s = \mathbb{I}^{\otimes n},$$

where $n = 2L^2$ is the total number of physical qubits (edges) in the lattice. Every qubit (link) appears in exactly two plaquette (or site) operators, so their contributions in the total product cancel out giving the above relation.

Due to this constraint, there are only $L^2 - 1$ independent plaquette and site operators respectively. Thus, this give us the total number of independent stabilizer generators by

$$r = 2(L^2 - 1).$$

C. Logical Codespace and Logical Operations

The commutation properties of the check operators, along with their independence allow us to define a set of stabilizer generators that form an Abelian subgroup of the Pauli group. Using the stabilizer formalism, we can find the number of encoded logical qubits k , given by

$$k = n - r$$

where, for an $L \times L$ toric code, $n = 2L^2$ is the number of physical qubits and $r = 2(L^2 - 1)$ is the number of independent stabilizers. The toric code can thus encode $k = 2L^2 - 2(L^2 - 1) = 2$ qubits.

The logical codespace is the $+1$ eigenspace of all the check operators. From our above analysis of the properties of the codespace, we can write the 2-qubit logical codespace as

$$|\psi_0\rangle = |\psi_L^{(1)}\rangle \otimes |\psi_L^{(2)}\rangle$$

which satisfies

$$A_s |\psi_0\rangle = |\psi_0\rangle \quad \text{and} \quad B_p |\psi_0\rangle = |\psi_0\rangle \quad \forall s, p$$

Alternatively, this can be understood as the absolute ground state of the Hamiltonian:

$$H = - \sum_s A_s - \sum_p B_p$$

To understand the logical operations on a torus, we begin by analyzing the different kinds of *cycles* on the lattice. By cycles, we mean a set of X or Z gates along any closed chain on the torus. ("closed" here is very loosely worded to mean the chain is a link that starts and ends at the same point. A more rigorous definition using homology theory is offered in [1], but I believe the underlying point is better conveyed in this intuitive way.)

Before presenting the analysis, it is fruitful to introduce the concept of duality between X and Z operations (or, equivalently, between plaquette and site operations) in our formalism. All analysis from here onward will be done solely with plaquettes (i.e., Z -type operations) in mind, and site operators will only be introduced whenever necessary. We claim that the same analysis applies to sites (i.e., X -type operations) due to the inherent duality of the lattice structure. Given a physical set of regularly spaced qubits arranged along the edges (or links) of a square lattice, we can define two different sets of lattice points (each shifted by half the lattice spacing in both directions) : (Figure 2):

- The **Primal lattice**, where Z gates correspond to plaquette operators.

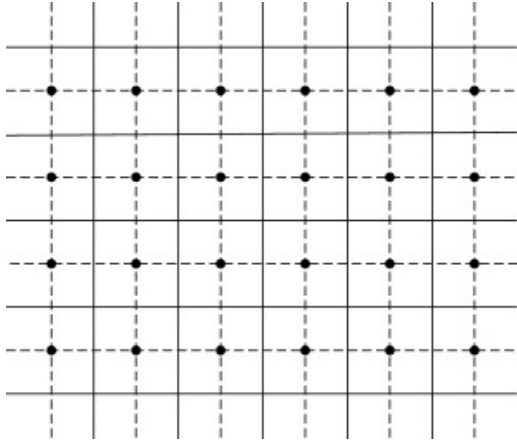


Figure 2. Primal and Dual lattices. Primal lattices are shown by dark lines whose intersections form the sites. The dual lattice is shown using dotted lines, and its sites are marked with black dots. They are related via a direct translation in $\frac{a}{2}\hat{x} + \frac{a}{2}\hat{y}$ direction, where a is the lattice spacing. [2]

- The **Dual lattice**, where Z gates correspond to site operators (and hence X gates become plaquette operators).

Therefore, any analysis done for Z -operations on the primal lattice applies equally to X -operations on the dual lattice, rendering it redundant to discuss both cases explicitly. Unless mentioned otherwise, all our analysis will be over the primal lattice, with plaquette operations referring to Z -stabilizers and site operations referring to X -stabilizers.

We now consider two different types of chains. As noted above, in the primal lattice a chain is a set of Z operations forming a loop (similarly with dual lattice/ X operations). Had we considered a chain as a continuous loop on a torus, we could've imagined it to be of two different classes (aka *homology*):

- A **homologically trivial** loop: one that can be shrunk to a point.
- A **homologically non-trivial** loop: one that wraps around the torus and cannot be shrunk to a point.

We extend this notion to our tiled lattice as follows. A homologically trivial cycle is one whose interior can be fully tiled by a set of plaquette operations. Such a cycle can be thought of as a chain of Z -gates forming a closed loop, which has the same effect as the product of all plaquette operators in its interior (since each interior link appears twice and cancels out, only the Z operations on the boundary contribute).

Therefore, this operation commutes with the codespace stabilizers and is also generated by it, as it

is generated by the set of check (stabilizer) operators. Hence, it acts trivially on the codespace.

On the other hand, a homologically non-trivial cycle is a chain of Z -operations that goes around the torus and returns to the same point. This chain does not bound any region, and hence is not generated by them, but, being composed entirely of Z -gates and forming a closed loop, it still commutes with all the check operators.

Such an operator belongs to the set defined as

$$\mathcal{N}(\mathcal{S}) \setminus \mathcal{S}$$

where $\mathcal{N}$ is the Normalizer of the stabilizer group $\mathcal{S}$, containing every operator that commutes with the generators of $\mathcal{S}$.

Hence, these operations preserve the codespace structure (because they commute with the stabilizers) but need not preserve the state within the codespace, thus forming perfect candidates for logical operations.

We make another simplification in our analysis by considering only one logical qubit (instead of the two that we previously found to exist). The intuition for this follows from the analysis of homologically non-trivial cycles.

We mentioned such a loop going around the torus, but in fact, there are two such loops on a torus (corresponding to two independent non-trivial cycles), which entail two different kinds of logical operations. We can show that these individual operators satisfy the single-qubit Pauli commutation relations.

Specifically, the logical operators $\bar{X}_i$ and $\bar{Z}_i$ (as shown in Figure 2) only overlap at one point, and hence they anticommute:

$$\{\bar{X}_i, \bar{Z}_i\} = 0.$$

Also, both square to identity:

$$\bar{X}_i^2 = \bar{Z}_i^2 = I,$$

which follows trivially from the Pauli nature of individual qubit operations. Therefore, we can arbitrarily choose one of the operations as logical $\bar{Z}_i$, which then renders the other as logical $\bar{X}_i$. $i = 1, 2$ labels the two logical qubits encoded by the two non-contractible cycles (i.e., the two distinct “turns” around the torus).

We can simply choose to discuss one set of operations for the rest of our analysis, and the arguments for the other Logical qubit can be made in a similar vein.

II. ERRORS IN THE TORIC CODE

A. Errors as defects on a lattice

The idea behind syndrome extraction highlights why *check* operators are named the way they are. In line

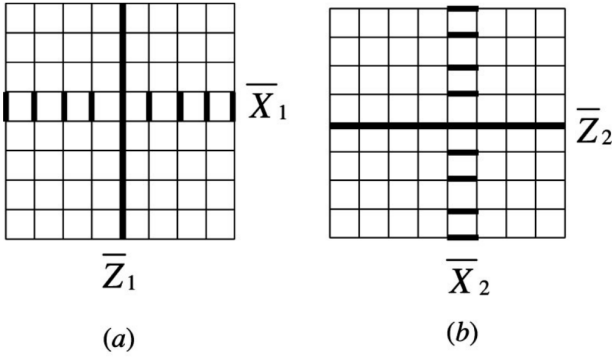


Figure 3. Logical gates on the two qubits of Toric code

with the stabilizer formalism, we use measurements of these operators (the relevant circuits are discussed in Appendix I) to check for errors on the physical qubits.

The local nature of these check operators allows us to isolate errors on a given “link”. For example, an X error on a single qubit manifests as a -1 parity measurement of the two Plaquette operators (each performing a $Z^{\otimes 4}$ measurement) that it belongs to. In the absence of errors, each check operator measurement returns $+1$, as per the construction of the logical state.

We can extend our analysis to understand bit-flip errors on multiple (or a chain of) qubits. Before doing that, it’s useful to introduce a picture where we assign each check operator measurement to a site on the Primal lattice. For the case of star measurements, this site is the vertex of the star, however, for Plaquette measurements we can consider the dual lattice where the same manifest as stars with obvious vertices, and hence have an obvious mapping to the points on dual lattice. We can then use the one-to-one translational correspondence of these two lattices to just assign any measurement on the qubits to a lattice site on the primal lattice.

We then observe that errors on a single qubit manifest as a -1 measurement in its nearest “sites” (where each lattice site has a mapping to a check operator of each type). For errors on adjacent qubits, the site connecting the two qubits no longer detects the error, as it acquires a factor of -1 from both thus returning to $+1$. As a result, the error is now detected only at the boundary vertices of these two edges (qubits).

This idea can be extended to the case of a chain of errors: we find that the syndrome manifests only at the end vertices of the chain. From the perspective of values of the check operators, these errors are fittingly referred to as **defects** on the lattice; These defects thus occur in pairs every time an error is incurred in a link. **As errors accumulate in a chain, two such defects tend to nucleate and propagate towards ends of**

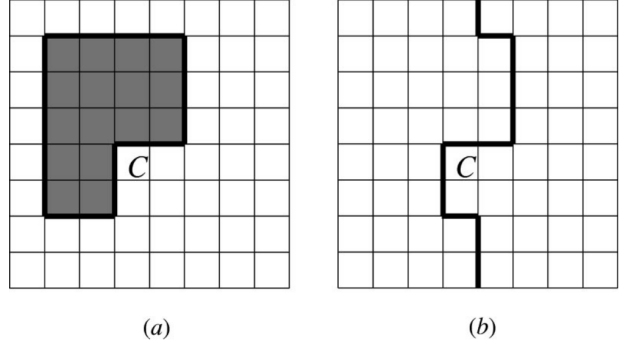


Figure 4. (a) Trivial and (b) non-trivial cycles on a torus

the chain.

B. Correctable vs Fatal errors

We see the importance of the discussion of cycles on a lattice when we try to understand the set of errors that are correctable (and, perhaps come up with a distance d for the code).

In a practical setting, every step of our computation involves a QEC round where we do the stabilizer measurement and update the values to the sites, and try to identify and correct for defects. In the simplest case, we find two defects on the lattice, the origin of which may be unknown (it may have occurred from any chain on the lattice that connects the two points). It turns out we do not need this information at all for the recovery step. To understand this, we use the fact that, regardless of any true “Error cycle” C along which the error may have occurred, if we are able to identify a chain C_1 that connects these defects such that $C_1 \cup C$ forms a closed loop (that can be shrunk to a point, aka *homologically trivial*), then we’ve shown that together they form the boundary of a region that is generated by the check operators, thus acting as the identity.

The major caveat here lies in identifying a C_1 that has the same *homology* as the error chain C : in a trivial case, both C and C_1 form a closed loop and thus the error is correctable. However, if the homology of C is wrongly identified – which would be the case where C goes around the torus once whereas C_1 does not – then our effective operation along $C_1 \cdot C$ forms a non-trivial cycle on the torus, and does not protect codespace information. These constitute the set of errors that are uncorrectable (in fact, this was our basis for looking at logical operations on the codespace!).

C. Distance of the Code

With the above analysis, we can now define the distance of our code. The distance d of a quantum code is defined as the minimum weight of a Pauli error string that does not preserve the encoded information. More formally,

$$d = \min \{ \text{wt}(E) \mid E \in \mathcal{P}_n, E \notin \mathcal{S}, [E, \mathcal{S}] = 0, \}$$

where $\mathcal{P}_n$ is the n -qubit Pauli group, $\mathcal{S}$ is the stabilizer group, and $\text{wt}(E)$ denotes the weight (number of non-identity components) of the Pauli string E .

We can argue that for an error to act non-trivially on the code space, it must form a loop that wraps around the torus at least once. Otherwise, any error can, in principle, be corrected by identifying the homology of the error chain connecting the defects.

An error chain E that wraps around the torus and forms a closed loop will not create any defects, as the endpoints cancel out (the defects can be viewed as being de-nucleated at the starting point of the closed chain). Such an error is topologically non-trivial and corresponds exactly to a logical operator (either a logical $\bar{Z}$ or $\bar{X}$, for the two different types of non-trivial loops). Hence, the minimal weight Pauli string that can implement a logical operation is the one that wraps around the torus in the shortest possible way.

For a toric code defined on an $L \times L$ square lattice, the shortest such non-contractible loop has length L . Therefore, the distance of the code is:

$$d = L$$

This implies that any error chain E with $\text{wt}(E) < d$ is detectable, which also directly follows from our analysis.

Thus, it is clear that the problem of error correction is reduced to estimating homology of the cycles, given a matrix of defects on the lattice. In the following section, an Error model is developed that helps with estimating the threshold probability for our code.

III. ERROR RECOVERY MODEL

Is syndrome detection sufficient for protecting QI for practical applications?

1. Practical errors and the Time-Space grid

Consider a situation where syndrome measurement is performed steps of dt , and during each time step there exists an IID probability distribution among all qubits

for incurring a X , Z error (given by p). Similarly, faulty measurements may also cause errors, given with a probability q . Each of these errors are assumed to be uncorrelated. Hence during each time step dt , the system can be modelled as a quantum channel that evolves the density matrix ρ as below:

$$\rho \rightarrow (1-p)^2 I \rho I + p(1-p) X \rho X + p(1-p) Z \rho Z + p^2 Y \rho Y$$

In any practical scenario, both p and q are expected to be non-zero. In our analysis on error propagation on lattice in the previous section, we had assumed that all syndrome measurements are perfect and correctly reflect the defects in the system. However, in presence of non-zero q , we expect an additional set of errors to be introduced due to wrongful reporting of site defect values – this can either be of the form of *ghost* defects (defects arising where there aren't really any) or not reflecting certain defects. (This also is effectively of bit-flip (or phase-flip) nature!) We expect to catch the errors that haven't been detected with repeated measurements, but, this imposes a tradeoff because it also leads to the number of wrongly reported ghost defects.

To study the effectiveness of the code in presence of such errors, we formulate the problem at hand as follows:

Does the syndrome information enable us to recover from errors with a probability of failure that becomes exponentially small as the size of the toric block increases?

We essentially want to be convinced that it is indeed fruitful from a practical perspective to scale up the code to a larger distance in order to effectively minimize errors to what is required.

Our analysis begins with presenting an equivalence between measurement errors in time and defects in space. To understand this, we study the example of a 1D version of a toric code (i.e. a repetition code on a circle, i.e. with periodic boundary condition). The correspondence of this 1D code with the 2D toric code is direct:

- an edge in the 1D code maps to a square in the 2D code
- a stabilizer measurement of the 1D repetition code ZZ map to plaquette measurement $ZZZZ$ in 2D
- errors manifest in sites at the ends of each link (qubit) the same way in 1D and 2D codes.

We now construct a 2D grid where horizontal axis corresponds to space (the 1D chain of atoms and array of lattice sites) whereas the vertical axis corresponds to discrete steps in time $t_n \rightarrow t_{n+1}$ between which one

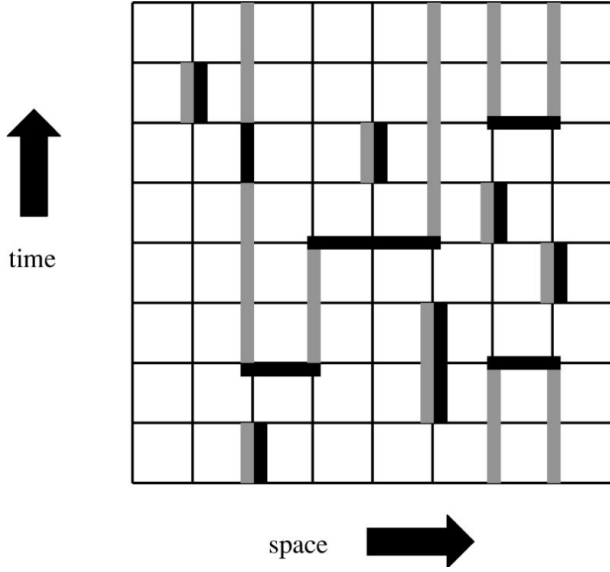


Figure 5. Error history map for a 1D Repetition code, where dark lines correspond to links in E and light vertical lines correspond to links in S , which is typically obtained from repeated stabilizer measurements)

syndrome measurement cycle is completed. Errors on the qubits at a given time step t_n appear as horizontal links, whereas those caused due to imperfect measurement during the interval $[t_n, t_{n+1})$ form vertical links between the lattice points at site s .

Hence, errors on horizontal links occur with probability p , and errors on vertical links occur with probability q .

Developing the lattice over several rounds of syndrome measurement, i.e., multiple steps in time results in an *error history map* that captures both the error events affecting the physical qubits in the code block, and faults arising in the syndrome measurements. An example error history map (Fig. 5) shows non-trivial (i.e. syndrome measurement $ZZ \rightarrow -1$) over several rounds of such measurements. Note that Syndrome history just gives a set of vertical lines on this plot. For our analysis, we will try to understand how this might relate to the true error history of the qubits.

2. Properties of error chains on the Time-Space grid

Consider a set of edges (not necessarily connected) S , the *syndrome chain*, consisting of all vertical links at which the measured syndrome is non-trivial (i.e., $X_s, Z_P = -1$).

Let E denote the *error chain*, which includes all links where errors have occurred. This includes both qubit errors on horizontal links and measurement errors on

vertical links. (Fig. 5 also shows an example of an error history map containing the two types of chains)

We can interpret S as the set of syndromes links on this world line (that are detected in practice over time), and E as the actual set of errors that may have occurred on the qubits (and during imperfect measurements) over time.

Let's define $+$ as the binary addition operation between the two sets, such that $S + E$ returns 1 if and only if an element $A_i \in S$ and $A_i \notin E$, or vice versa:

$$T = S + E = (S \cup E) \setminus (S \cap E)$$

where we have defined T to be the true set of defect links. $+$ is addition modulo 2, so if an error is in both S and E , it gets removed from $S + E$. This implies $T = S + E$ is actually set of all errors that are actually *not* corrected for (because those errors in S that were not corrected from imperfect measurements as errors in E). Hence it represents the **actual defect worldline** generated by all errors in the system, after accounting for all kinds of errors including those corrected for by our faulty syndrome recovery procedure.

The reason this set of links T is significant is because it follows an important property: For a measurement that occurs repeatedly over infinite time, these world lines do not end, i.e. the chain of links T do not have a boundary. This is closely related to the argument for a 2D Toric code about errors being created in pairs. We can argue that this should be true by considering the following situation:

Assume no errors exist, and the first error occurs as a vertical link which is detected as a faulty syndrome (i.e. a syndrome error is detected where there isn't any). This type of error doesn't manifest in the set T because such an error would belong to both E and S . However, an error on a horizontal link joining sites $s \rightarrow s+1$ would immediately reflect in our set S as vertical lines on sites s and $s+1$ in the next time step (if at all measurement is not faulty). In case of a faulty measurement, we do not see such an error in our set S , but since the measurement is faulty it will appear in E , because of which it would still appear in T . Hence this error propagates in time in a closed manner, and as the qubit errors nucleate the width of the cycle increases (or decreases, in the extreme case the defects on sites $s, s+1$ both nucleate and we end up with a closed loop. Since measurements are considered infinite in time, these world lines do not terminate, and hence there doesn't exist a boundary vertex for the same. This condition can be written as,

$$\delta(T) = \delta(S + E) = 0 \Rightarrow \delta S = \delta E$$

implying that both sets S and E share the same boundary.

The consequence of the above condition is that, we are able to treat S and E a similar way we did two chains with no boundary. A measured syndrome S thus also reveals boundary of the true error chain E , given by δE . Hence, there exists a cycle C on the time-space grid such that $E = S + C$. For recovery, we look at the syndrome map S and would like to predict a cycle C which would have caused the error E . Since there may exist more than one such cycle C' that give rise to an error $E' = S + C'$ which have the same boundary as S (i.e. more than one chain of error E exists which give same syndrome as S), our task for recovery is now boiled down to predicting a cycle C' , such that,

$$E + E' = C + C' = (S + C) + (S + C')$$

i.e. if given a set S , if we are able to predict a chain E' such that $E + E' = C + C'$ is a homologically trivial cycle. We use the result that if $C + C'$ homologically trivial, then C, C' **individually must belong to the same homology class (i.e. either both are trivial or non-trivial)**, and hence claim that if we are able to successfully identify the *homology class* of cycles C, C' then recovery will be successful. Thus, we build our mathematical model to derive the probability of rightly identifying the homology class of cycle C , which indirectly relates to our quest for the threshold probability.

3. Mathematical Model and Threshold Probability

We want to now relate these ideas to threshold probability of single-qubit errors on a qubit, that guarantees recovery in some limit. Since we have reduced the problem of recovery down to finding the homology classes of the cycle $D = C + C'$, we can impose conditions for a successful recovery as follows:

For a cycle D such that $E = E' + D$ where E is the true error chain and E' is the predicted chain, we impose that:

$$\sum_E p(E) \cdot \sum_{\substack{D \\ \text{nontrivial}}} p[(E + D) | E] = 0.$$

above equation should hold in the limit of large code block, i.e. $L \rightarrow \infty$ for us to be able to achieve error per qubit $p < p_{th}$, which is the threshold probability. This is an important result because it uses the idea of a mathematically derived threshold p_{th} and imposes a physical criterion that argues that for larger and larger encoding to be successful, we must expect the probability of our hypothetical chain E' resulting in a non-trivial cycle D to be zero for all possible error history chains E .

We can make the above model physical by actually deriving the probabilities. The probability that an error

chain E' occurs is given by,

$$p(E') = \prod_{\ell} (1 - p)^{1 - n_{E'}(\ell)} p^{n_{E'}(\ell)} \sim \prod_{\ell} \left(\frac{p}{1 - p} \right)^{n_{E'}(\ell)},$$

where the occupancy function $n_{E'}(\ell) = 1$ iff link $\ell \in E'$ and is 0 otherwise. We can similarly write $E = E' + C$ and an occupancy function for cycle C as $n_C(\ell)$ with a similar definition. For a given error chain E , we want to predict the probability that the chain $E' = E + C$ has same boundary as E (i.e. C is a cycle). This can be calculated as follows:

- The probability that a link ℓ lies in E' but not in E is equivalent to the probability that ℓ belongs to the cycle C .

Specifically, if $n_E(\ell) = 0$, then from the relation $E' = E + C$, it follows that $n_{E'}(\ell) = n_C(\ell)$. Therefore, the probability that link ℓ belongs to E' (and not E) is the same as the probability that ℓ belongs to C , which is:

$$p^{n_C(\ell)} (1 - p)^{1 - n_C(\ell)}.$$

- Similarly, if a link ℓ lies in E but not in E' , then $n_E(\ell) = 1$, and since $n_{E'}(\ell) = n_E(\ell) + n_C(\ell) \bmod 2$, we have $n_{E'}(\ell) = 1 - n_C(\ell)$. This occurs with a probability given by

$$p^{1 - n_C} (1 - p)^{n_C} \sim \left(\frac{1 - p}{p} \right)^{n_C}$$

Hence, probability that a chain E' having the same boundary as E (i.e. satisfies the above conditions) is given by:

$$p(E' | E) = \prod_l \exp(J_l u_l)$$

where, $u_l = 1 - 2n_C(l)$ is an Ising variable, and l runs over all lattice sites and is of two kinds ($l \in E, l \notin E'$ and vice versa (as discussed above) and thus gives us a piece-wise definition of J_l as

$$e^{-2J_l} = \begin{cases} \frac{p}{1 - p}, & l \notin E \\ \frac{1 - p}{p}, & l \in E \end{cases}$$

To ensure u_l represents our cycle, we impose a condition that set of links l given by $u_l = -1$ (i.e. $n_C(l) = 1$) must always form a cycle. This is done in 2D by the following constraint:

$$\prod_{l^* \in P^*} u_{l^*} = 1$$

for all Plaquettes P^* in dual lattice. (The constraint says that every cycle must pass through at least two links on a Plaquette, or in an equivalent 3D version containing one time dimension and two spatial dimensions (of the 2D Toric code), that for every cycle, the function u_P must satisfy (now, links have mapped to plaquettes and plaquettes to cubes on the 3D lattice:)

$$\prod_{P^* \in C^*} u_{P^*} = 1$$

A solution for the 2D system (one time-like and one space-like axis) involves associating an Ising spin variable σ_{l^*} to every link that makes up the 2D Plaquette and writing $u_{l^*} = \prod_{l^*} \sigma_{l^*}$. The probability has hence been shown to be equal to the partition function of a 2D Ising model, given by

$$Z[J, \eta] = \sum_{\{\sigma_i\}} \exp \left(J \sum_{\langle ij \rangle} \eta_{ij} \sigma_i \sigma_j \right),$$

where the summation $\langle ij \rangle$ is over the nearest neighbours to σ_i and the piece-wise function defined previously is separated out into a constant $e^{-2J} = \frac{p}{1-p}$ and η which takes values $+1, -1$ for $l \notin E^*, l \in E^*$ respectively. This is a common model studied in statistical physics, known as the *random-bond Ising model* followed by disordered spin systems [3]. It derives its name from the statistical treatment used to study the model, where η_{ij} is sampled from some probability distribution (hence the random-bond).

The key takeaway from the above exercise is that, mapping the probability to a partition function of an Ising spin system allows us to correlate the threshold probability in our system p_{th} with a critical point in the random-bond Ising model. In the analysis of the Ising model, there exists a phase transition between "ordered" and "disordered" phases, and the critical point is evaluated as the intersection point of this phase boundary with Nishomori line defined by $e^{-2J} = \frac{p}{1-p}$.

The ordered phase can be interpreted as one where error chains do not extend far, and hence the cycles C relating E, E' are local (making the errors correctable), and hence making the quantum memory reliable. On the other hand, "disordered phase" is more susceptible to logical errors and hence the code is not effective in correcting the syndromes.

It was also shown that in the limit $L \rightarrow \infty$, the spin correlations (which act as a "Disorder parameter" in the Ising model) decay with distance and effectively go to zero in our limit, which directly maps back to our criterion of probability of non-trivial cycles going to zero.

The phase diagram is shown for Random-Bond Ising Model in Fig 9. that shows the intersection of Nishomori line with the phase diagram plot and the value of threshold probability is thus obtained as,

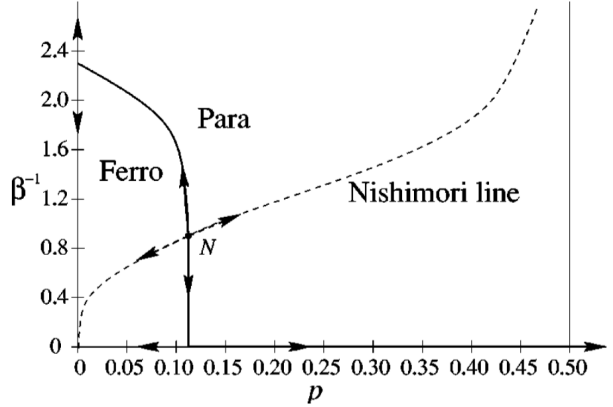


Figure 6. The phase diagram of the random-bond Ising model. Vertical axis is directly proportional to coupling strength J and p is the probability of "antiferromagnetic bond" $\eta_{ij} = -1$ on the horizontal axis. The phase boundary and its crossing with the Nishomori line at the Nishomori point N is plotted, which gives the critical (threshold) probability for the order-disorder transition.

$$p_c \approx 0.1094$$

IV. CONCLUSION

In this report, we have provided a detailed analysis of the Toric code using Stabilizer framework and highlighted the existence of a logical qubit as well as the performing logical operations on them. We have also presented a detailed analysis of how errors manifest during syndrome measurements, and the distinct advantage of check operators as stabilizers in helping to localize the errors. We further looked into syndrome measurements for performing recovery operations, and analyzed the situations that were fatal to the logical quantum information. Finally, we rigorously developed a framework that helps establish the threshold probability for our system, which is crucial to understand from the perspective of practical usage of these codes. In doing so, we have analyzed the types of errors, both due to direct qubits (that model phenomenon like decoherence), as well as errors during the stabilizer measurements. We imposed a constraint which necessitated that the probability that different error chains share the same homology class, and found that this mapped to the problem of random-bond Ising model in statistical mechanics which studies disordered systems, has a phase boundary with a critical point w.r.t the probability parameter p that also gives the threshold probability for our system as $p_{th} \approx 0.1094$.

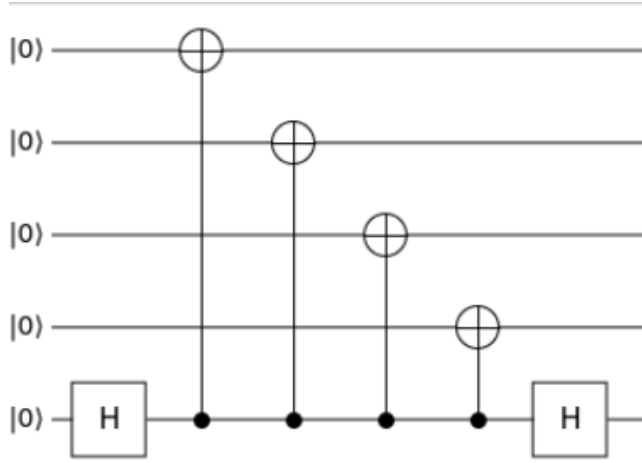


Figure 7. Circuit for X_s measurement. The top 4 are the data qubits and the bottom one is the Ancilla. [4]

Although much of our analysis used simplifying assumptions (such as measurement error and single-qubit error share the same probability, $q = p$, $t_n \rightarrow \infty$, ...) a lot of further analysis is pursued in [1] which highlights that the results are not so different when these assumptions are relaxed. The qualitative behaviour of the model remains prevalent throughout different scenarios. This fundamentally can be thought of as arising from the parallels of the behaviour of defects in our system with that of the magnetic monopole arising in disordered spin systems, that is likely to result in multiple qualitative similarities inspite of our assumptions.

Appendix: Initializing (Encoding) the Logical State

Assuming perfect measurements, the following procedure can be used to encode qubits in the logical state

$|0\rangle_L$:

1. Initialize every physical qubit to the state $|0\rangle$, i.e., the initial state is $|0\rangle^{\otimes n}$.
2. This state is a simultaneous $+1$ eigenstate of all plaquette operators Z_p , but not of the site (star) operators X_s .
3. "Measure" all site (star) operators X_s . Since these commute with the plaquette operators, the measurement preserves the plaquette eigenvalues and thus the state remains in the $+1$ eigenspace of all Z_p .
4. Each site operator measurement (Figure 7.) yields ± 1 with equal probability. This is because each qubit is in the $|0\rangle$ state, which can be expressed in the X -basis as

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle),$$

so that for n qubits,

$$|0\rangle^{\otimes n} = \left(\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) \right)^{\otimes n}.$$

Measuring each qubit in the X -basis gives ± 1 outcomes with equal probability, and hence each site operator, being a product of several X 's also yields ± 1 outcomes randomly.

Now, consider each site with a -1 result as a defect and apply standard recovery procedure, thus bringing the initial state to a $+1$ eigenstate of all check operations.

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